

TEAM CIPHER · 2-DAY WORKSHOP

Flutter Fundamentals

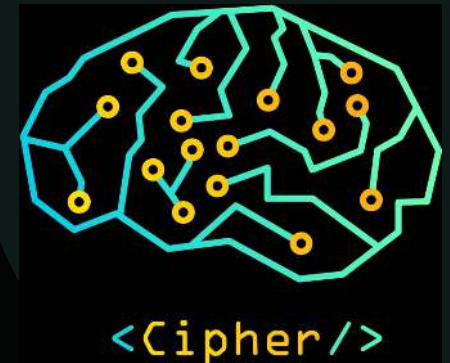
Build Your First App — from Flutter Widgets to a Working Counter App

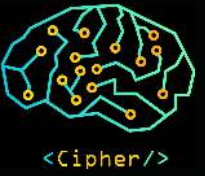
Instructor

Alaa Elnagih

Day 1 · Flutter

Day 2 · State & UI





Day 2

State, StatefulWidget, and a Real Counter App

QUICK REVISION

StatelessWidget vs StatefulWidget

StatelessWidget

- Fixed once drawn
- Example: logo, static text
- No setState needed

Yesterday's Profile Card

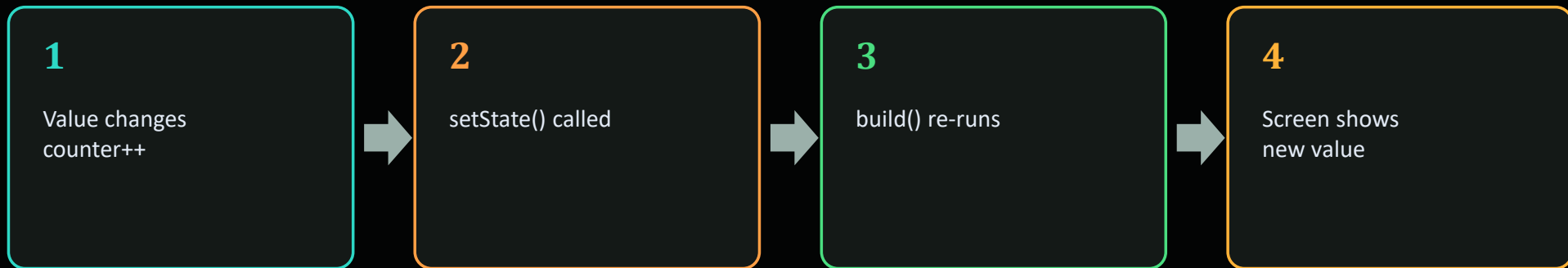
StatefulWidget

- Can change while visible
- Example: a tap counter
- Needs setState to redraw

Today's Counter App

THE CORE MECHANIC

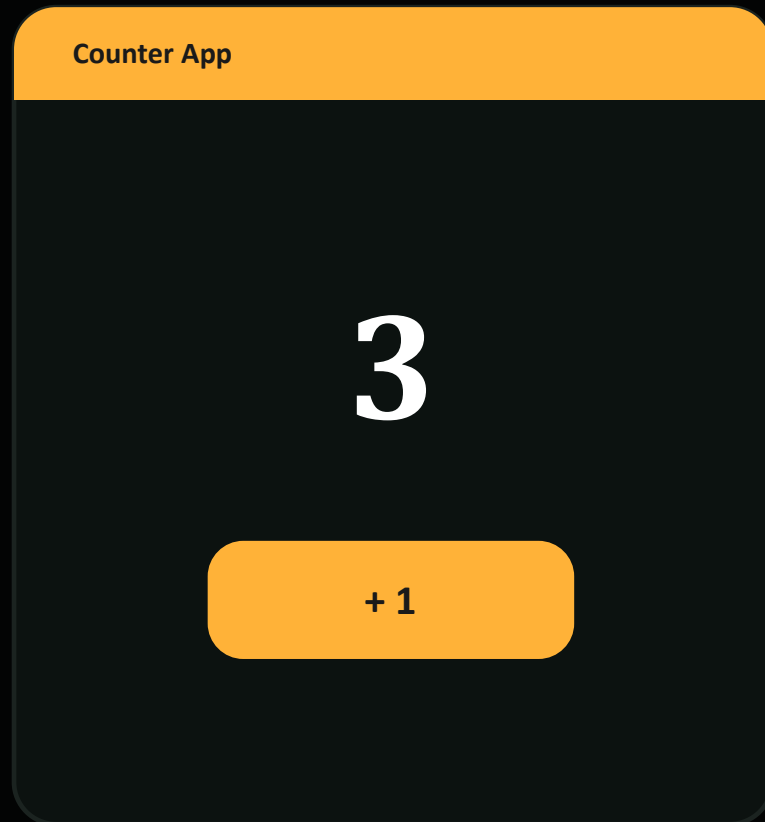
setState — Update the Value, Redraw the Screen



```
void increment() {  
  setState(() {  
    counter++; // change the value AND ask Flutter to redraw  
  });  
}
```

Without `setState`, the value changes in memory — the screen never finds out.

Building It: Simple Counter



One screen, one number

- One State variable holds the count: a single int, starting at 0.
- ONE button — ElevatedButton with a Text('+ 1') child.
- Tapping it calls one function that adds 1 and calls setState.
- setState tells Flutter to re-run build() and show the new number.

One Variable, One Button

```
void increment() {  
  setState(() {  
    counter++;  
  });  
}  
  
// counter starts at 0  
int counter = 0;
```

ONE function, ONE job: add 1 to counter, then tell Flutter to redraw.

```
Text('$counter', style: ...  
    fontSize: 60)),  
  
ElevatedButton(  
  onPressed: increment,  
  child: Text('+ 1'),  
)
```

The button's onPressed just points at increment — no arguments needed.

WORKSHOP RECAP

You Built a Real Flutter App — What's Next?



1

Navigation

Multi-screen apps, passing data

2

Flutter UI

More widgets, animations, theming

3

State Management

Provider, Riverpod, Bloc

4

Data

APIs, local storage, Firebase

5

Real Projects

Publish to Play Store / App Store

Thank you!

Team Cipher · Flutter Fundamentals